

Ashmita Dua

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Education

University of California, Santa Cruz | Undergraduate
B.Sc. Computer Science: Game Design

Sept, 2024 - June, 2028

Skills / Tech stacks

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|---------------------------------------|--------------------|-----------------------------|
| • LLMs/ RAG | • NNs/CNNs/Pytorch | • LangChain |
| • Procedural Content Generation (PCG) | • OpenCV | • JavaScript |
| • HTML/CSS | • Git, APIs | • Fluent in English & Hindi |
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Experience

Augmented Design Lab
Undergraduate Researcher

September, 2025 - Present

- Independent study focused on generating game levels through LLMs\PCG.
- Working with Procedural content generation, through hierarchical wave function collapse.
- Worked on **Pewter Platformer**, More in Projects section

Undergraduate Admissions Office, UCSC
Website Designer

January, 2025 - Present

- Designed, edited, and maintained the company's website, ensuring a user-friendly interface and up-to-date content.
- Managed and updated digital records, organizing and maintaining structured data for improved accessibility and efficiency.

Artificial Intelligence Explainability Accountability Lab
Undergraduate Researcher

April, 2025 - June, 2025

- Independent study focused on LLM safety, logic translation, and inconsistency detection.
 - Contributing to the LLM Logic project by setting up local environments (LLM API, SWI-Prolog).
 - Building a knowledge base and curating UCSC-specific questions.
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Projects

Pewter Platformer | Demo: pewterplatformer.wessel.xyz/

- Diagnosed an LLM failure mode where the agent entered repetitive validation loops during level creation
- Iterated the system prompt + interaction flow to enforce "action-first" behavior and structured tool use
- Implemented a lightweight conversation-state guardrail to limit repeated follow-up questions and reset on tool calls
- Benchmarked over 30 test prompts: loop-rate and avg questions/session 78.7% → 46.3%

Woven Mood-Based Storytelling LLM | Demo: duashmita.github.io/woven/ | Repo: bit.ly/woven-intoyou

- Developed a mood-guided LLM agent using system prompts + structured outputs to generate personalized interactive narratives.
- Implemented session state + memory (story variables, user preferences, summaries) to maintain coherence in multi-turn sessions.
- Integrated RAG (story rules / context retrieval) + output validation to reduce contradictions; demo UI prototyped with Lovable.

ASL Decoder CNN Model | Demo: duashmita.github.io/sign-language-vision/ | Repo: bit.ly/ASLDecoder-CNN

- Designed a CNN-based system for real-time American Sign Language (ASL) recognition using live video inputs.
- Applied **data augmentation** and optimized CNN architecture to enhance accuracy.
- Evaluated on webcam dataset to quantify domain shift; improved robustness via augmentation.